

EVIDENCE PRESERVATION AND EVIDENCE COLLECTION LAB

1. Objective

The objective of this lab is to perform volatile evidence collection, memory acquisition, evidence hashing, and chain-of-custody documentation using forensic tools including Velociraptor, FTK Imager, and sha256sum.

2. Tools Used

Tool	Purpose
Velociraptor	Network and memory evidence collection
FTK Imager	Memory capture and forensic imaging
Kali Linux	Evidence hashing
sha256sum	Integrity verification

3. Lab Environment

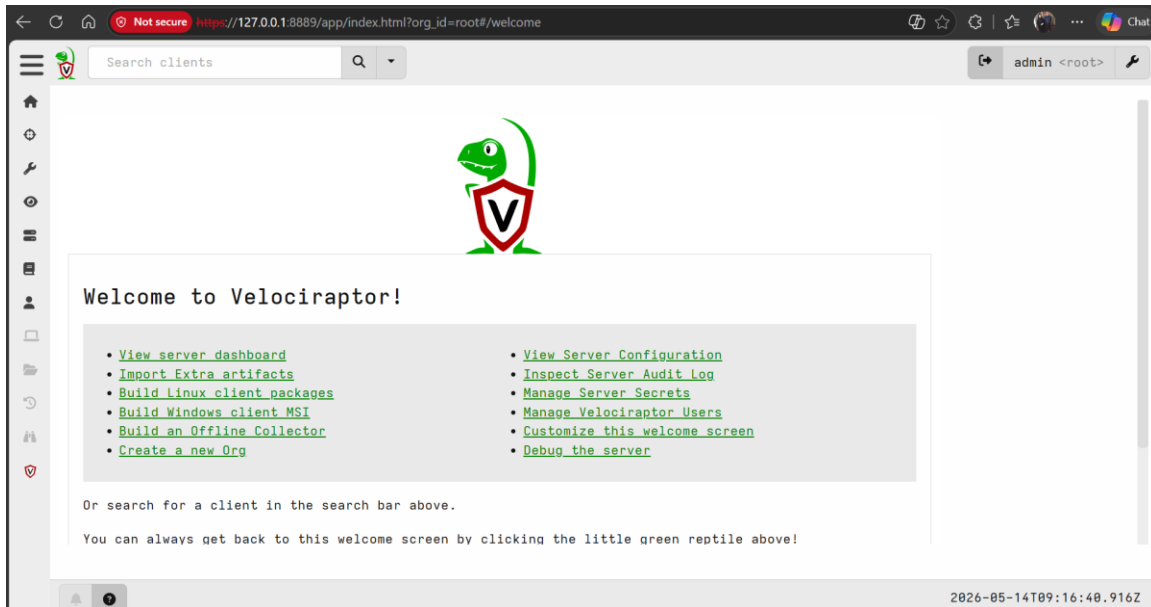
Component	Details
Windows VM	Evidence collection target
Kali Linux	Hash generation
Velociraptor GUI	Artifact collection
FTK Imager	Memory acquisition

4. Volatile Data Collection using Velociraptor

The Velociraptor DFIR platform was used to collect volatile network connection evidence from the Windows virtual machine. The netstat artifact query was executed to identify active network connections, remote IP addresses, ports, and process identifiers.

- Query Used:

SELECT * FROM netstat



velociraptor_dashboard.png

Administrator: Command Prompt				
Proto	Local Address	Foreign Address	State	PID
TCP	0.0.0.0:135	0.0.0.0:0	LISTENING	1904
TCP	0.0.0.0:445	0.0.0.0:0	LISTENING	4
TCP	0.0.0.0:5040	0.0.0.0:0	LISTENING	8708
TCP	0.0.0.0:49664	0.0.0.0:0	LISTENING	1608
TCP	0.0.0.0:49665	0.0.0.0:0	LISTENING	1440
TCP	0.0.0.0:49666	0.0.0.0:0	LISTENING	3328
TCP	0.0.0.0:49667	0.0.0.0:0	LISTENING	3500
TCP	0.0.0.0:49668	0.0.0.0:0	LISTENING	4796
TCP	0.0.0.0:49669	0.0.0.0:0	LISTENING	1580
TCP	127.0.0.1:8000	0.0.0.0:0	LISTENING	12464
TCP	127.0.0.1:8000	127.0.0.1:63032	ESTABLISHED	12464
TCP	127.0.0.1:8000	127.0.0.1:63035	ESTABLISHED	12464
TCP	127.0.0.1:8000	127.0.0.1:65364	ESTABLISHED	12464
TCP	127.0.0.1:8000	127.0.0.1:65366	ESTABLISHED	12464
TCP	127.0.0.1:8001	0.0.0.0:0	LISTENING	12464
TCP	127.0.0.1:8001	127.0.0.1:61547	ESTABLISHED	12464
TCP	127.0.0.1:8003	0.0.0.0:0	LISTENING	12464
TCP	127.0.0.1:8889	0.0.0.0:0	LISTENING	12464
TCP	127.0.0.1:8889	127.0.0.1:64190	ESTABLISHED	12464
TCP	127.0.0.1:61547	127.0.0.1:8001	ESTABLISHED	12464
TCP	127.0.0.1:63032	127.0.0.1:8000	ESTABLISHED	12464
TCP	127.0.0.1:63035	127.0.0.1:8000	ESTABLISHED	12464
TCP	127.0.0.1:64190	127.0.0.1:8889	ESTABLISHED	14656
TCP	127.0.0.1:65364	127.0.0.1:8000	ESTABLISHED	12464
TCP	127.0.0.1:65366	127.0.0.1:8000	ESTABLISHED	12464
TCP	192.168.1.40:139	0.0.0.0:0	LISTENING	4
TCP	192.168.1.40:49414	4.213.25.242:443	ESTABLISHED	5612
TCP	192.168.1.40:49670	13.232.193.230:443	ESTABLISHED	4428
TCP	192.168.1.40:50345	38.254.180.244:53	TIME_WAIT	0

netstat_query.png

State	FlowId	Artifacts	Created	Last Active	Creator	Mb	Rows
✓	F.D82PFRLNG7C06	Windows.Network.N etstat	2026-05-14T09:37:50.841Z	2026-05-14T09:37:51.792Z	admin	0 b	126
✓	F.D82P5NLDGV120	Generic.Client.In fo	2026-05-14T09:16:14.510Z	2026-05-14T09:16:44.158Z	InterrogationSe rvice	0 b	26

Pid	Name	Family	Type	Status	Laddr.IP	Laddr.Port	Raddr.IP	Raddr.Port
1984	svchost.exe	IPv4	TCP	LISTEN	0.0.0.0	135	0.0.0.0	0
4	System	IPv4	TCP	LISTEN	192.168.1.40	139	0.0.0.0	0
4	System	IPv4	TCP	LISTEN	192.168.56.1	139	0.0.0.0	0
8788	svchost.exe	IPv4	TCP	LISTEN	0.0.0.0	5040	0.0.0.0	0

netstat_results.png

The collected network evidence was exported into CSV format and securely stored inside the Evidence directory for preservation and future forensic analysis.

5. Memory Acquisition using Velociraptor

A full physical memory acquisition was performed using the Velociraptor memory acquisition artifact. The generated memory dump contains volatile artifacts such as running processes, network sessions, credentials, and malware traces.

- Query Used:

SELECT * FROM Artifact.Windows.Memory.Acquisition

The screenshot displays the Velociraptor web interface. The top bar shows the client name 'client', a search icon, and the connection status 'DESKTOP-J00T62L.hgu_lan Connected'. The sidebar on the left contains various navigation icons. The main content area is divided into two panels. The left panel shows the artifact details for 'Artifact.Windows.Memory.Acquisition', including Flow ID, Creator, Create Time, Start Time, Last Active, Duration, State, Error, and various limits. The right panel shows the 'Artifacts with Results' section, including Total Rows, Uploaded Bytes, Files uploaded, and a 'Download Results' button. A message 'Select a download method' is displayed below the download button.

Artifact Names	Windows.Memory.Acquisition
Flow ID	F.D82PTHG7QEHNQ
Creator	admin
Create Time	2 minutes, 7 seconds
Start Time	ago
Last Active	2026-05-14T10:07:48.369Z
Duration	46.01 seconds
State	Error
Error	Upload bytes 1074790400 exceeded limit 1073741824 for flow F.D82PTHG7QEHNQ. Cancelling.
Ops/Sec	Unlimited
CPU Limit	Unlimited
IOPS Limit	Unlimited
Timeout	600 seconds
Max Rows	1m rows
Max Mb	1024.00 Mb

Artifacts with Results

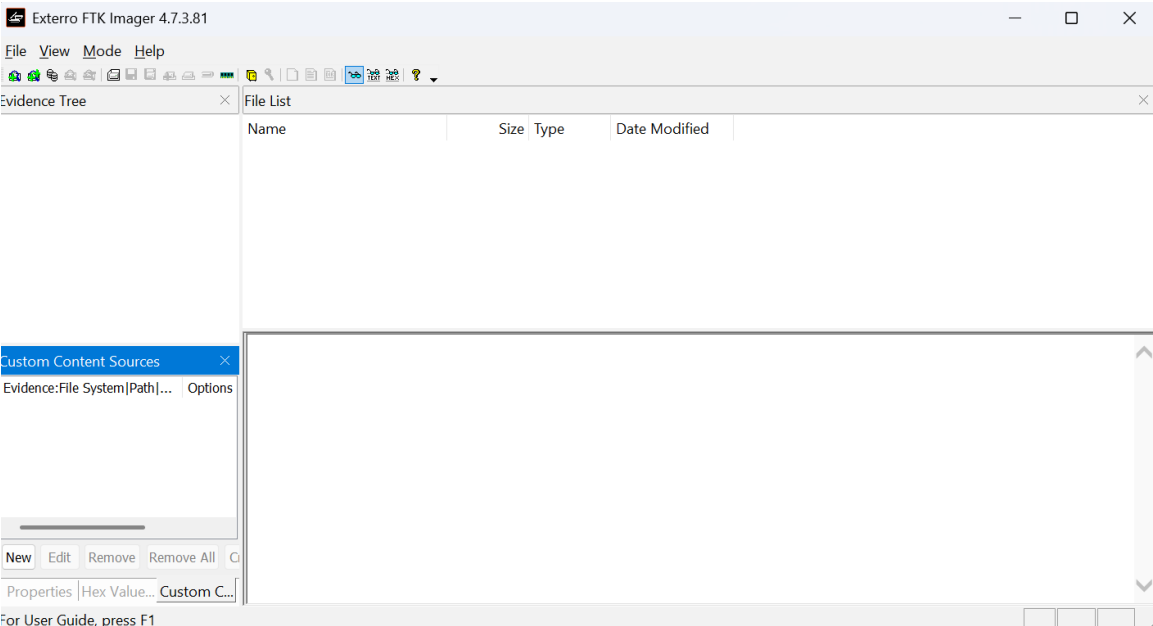
Total Rows	0
Uploaded Bytes	1074790400 / 17417895936
Files uploaded	1
Download Results	

Select a download method

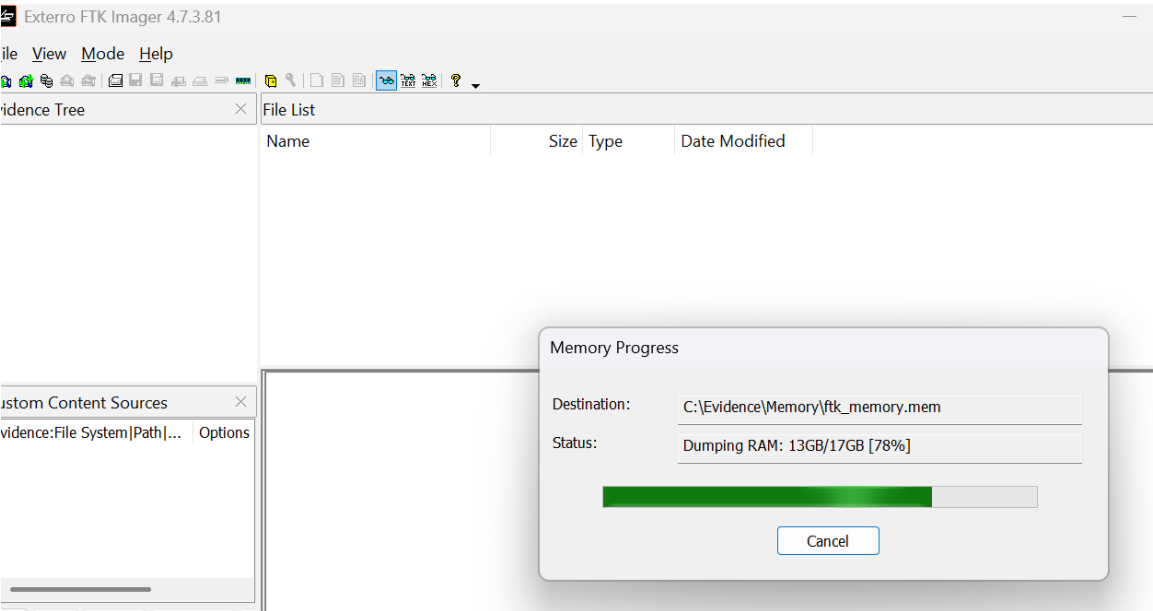
Memory acquisition artifact execution in Velociraptor.

6. Memory Capture using FTK Imager

FTK Imager was installed and used to perform memory capture operations and forensic evidence handling. The tool provides evidence acquisition and integrity verification capabilities.



ftk_dashboard.png



ftk_memory_capture.png

7. Evidence Integrity Verification using SHA256

The acquired memory dump was transferred to Kali Linux and hashed using the sha256sum utility to verify evidence integrity and detect unauthorized modifications.

- Commands Used:

```
sha256sum memorydump.raw
```

```
sha256sum memorydump.raw > hash.txt
```

```
sha256sum -c hash.txt
```

Algorithm	Hash	Path
SHA256	2F8BBE4F368C71D73F89BE9D89B02F2FCF181C2273D47F34401823D2C7D3CA83	C:\Evidence\Memory\memorydump...

sha256_hash.png

```
kali@kali:~/evidence$ sha256sum -c hash.txt
memorydump.raw: OK
```

hash_verification.png

8. Chain of Custody Documentation

Item	Description	Collected By	Date	Hash Value
Memory Dump	Windows VM RAM Dump	Faizan	2026-05-14	

Item	Description	Collected By	Date	Hash Value
Memory Dump	Windows VM RAM Dump	Faizan	2026-05-14	02EA66F993AF0
				AADCEDED129
				CB111570C196
				D8D98C5F1888
				1A9D4D0CEF014DA1

9. Conclusion

The lab successfully demonstrated evidence preservation and forensic acquisition techniques using Velociraptor, FTK Imager, and SHA256 hashing. Volatile network evidence and memory dumps were collected, preserved, hashed, and documented according to forensic handling procedures.

